

MATH-351 Numerical Method

Root Finding Problem

DBS&H, CEME-NUST

November 23, 2020

Outline

① Prerequisite

Equation of Straight Line

Taylor's Theorem/Series

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Equation of Straight Line

Name of the form	Equation	Given
Point-slope form	$y - y_1 = m(x - x_1)$	slope m , point (x_1, y_1)
Slope-intercept form	$y = mx + b$	slope m , b - y -intercept
Two-point form	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$	points (x_1, y_1) (x_2, y_2)
Intercept form	$\frac{x}{a} + \frac{y}{b} = 1$	a - x -intercept b - y -intercept
General form	$Ax + By + C = 0$	A and B are not both 0
Horizontal line	$y = b$	
Vertical line	$x = a$	

Equation of straight line passing through (x_1, y_1) and (x_2, y_2) is

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}.$$

Is the above equation is different from

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \quad ?$$

Outline

① Prerequisite

Equation of Straight Line

Taylor's Theorem/Series

The Taylor Theorem and Series

Taylor theorem The Taylor theorem states that any smooth function can be approximated as a polynomial.

Taylor series The Taylor series provides a means to express this idea mathematically.

$$f(x) = f(x_0) + (x - x_0) \frac{f'(x_0)}{1!} + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + R_n$$

In general, the n^{th} order Taylor series expansion will be exact for an n th order polynomial. In other cases, the remainder term R_n is of the order of h^{n+1} , meaning:

- The more terms are used, the smaller the error, and
- The smaller the spacing, the smaller the error for a given number of terms.

Newton-Raphson's Method

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$$f(x) - f(x_0) = f'(x_0)(x - x_0)$$

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If the tangent intersect x -axis
at $x = x_1$, then $f(x_1)$ is zero.

Thus

$$-f(x_0) = f'(x_0)(x_1 - x_0)$$

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$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

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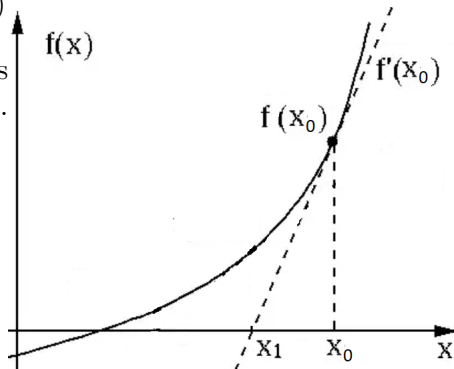
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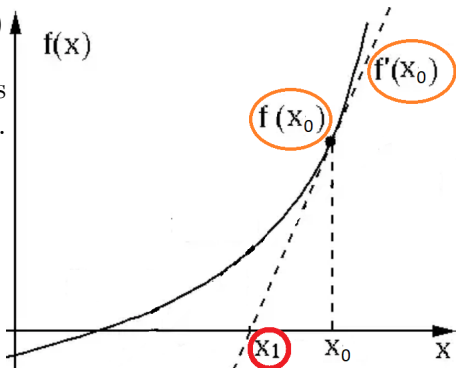
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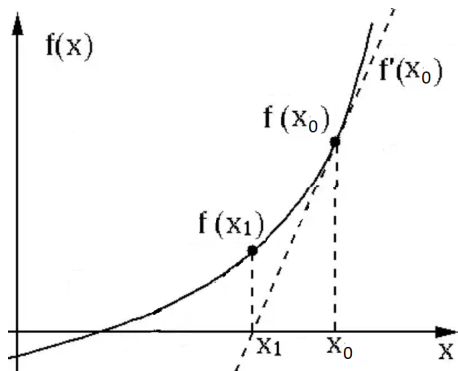
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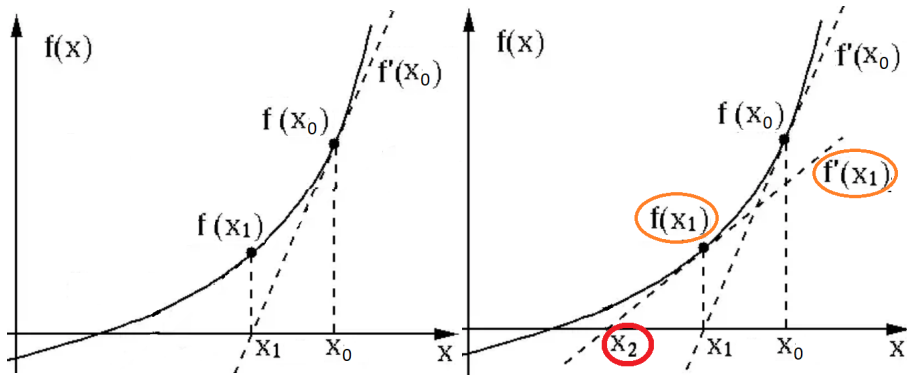
Newton-Raphson's Method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{provided } f'(x_1) \neq 0.$$

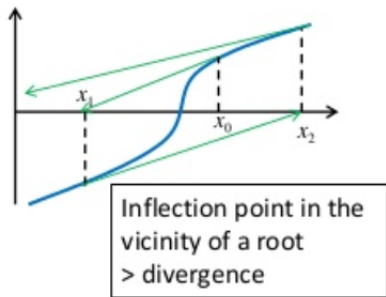


Newton-Raphson's Method

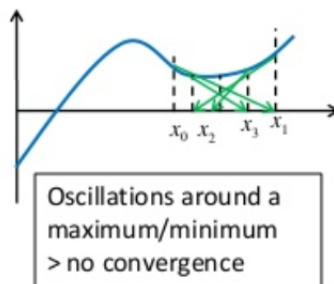
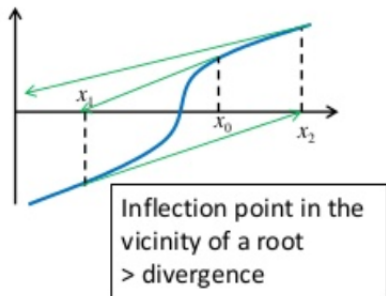
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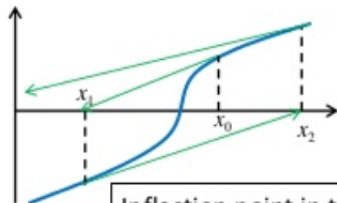
Pitfall of NR method



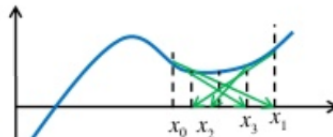
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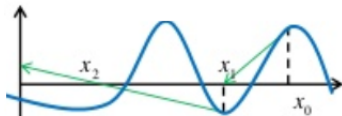
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Inflection point in the vicinity of a root
> divergence

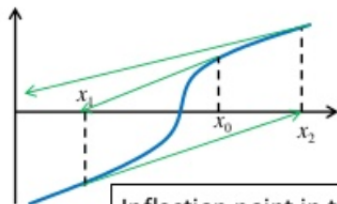


Oscillations around a maximum/minimum
> no convergence

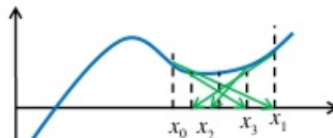


Initial guess close to one root
jumps to another root
> solution jumps away

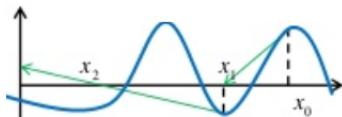
Pitfall of NR method



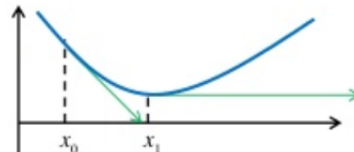
Inflection point in the vicinity of a root
> divergence



Oscillations around a maximum/minimum
> no convergence



Initial guess close to one root
jumps to another root
> solution jumps away



A zero slope is encountered
> Solution shoots-off

Determine the positive root of $f(x) = x^{10} - 1$ using the Newton-Raphson method and an initial guess of $x = 0.5$.

Secant Method

The equation of Secant line for the function $f(x)$ passing through $(x_0, f(x_0))$ and $(x_1, f(x_1))$ is given by

$$\frac{f(x) - f(x_1)}{x - x_1} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

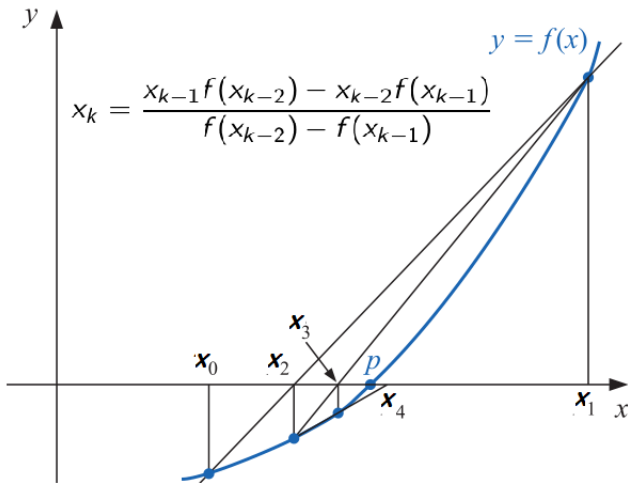
The secant line intersect the x -axis at $x = x_2$ assuming $f(x_2) \simeq 0$ gives

$$\begin{aligned} \frac{0 - f(x_1)}{x_2 - x_1} &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\ \frac{x_2 - x_1}{-f(x_1)} &= \frac{x_1 - x_0}{f(x_1) - f(x_0)} \\ x_2 &= x_1 - \frac{(x_1 - x_0) f(x_1)}{f(x_1) - f(x_0)} \end{aligned}$$

In general,

$$x_k = x_{k-1} - \frac{(x_{k-1} - x_{k-2}) f(x_{k-1})}{f(x_{k-1}) - f(x_{k-2})} \quad \text{or} \quad x_k = \frac{x_{k-1} f(x_{k-2}) - x_{k-2} f(x_{k-1})}{f(x_{k-1}) - f(x_{k-2})}$$

$$x_k = x_{k-1} - \frac{(x_{k-2} - x_{k-1}) f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})} \quad \text{or}$$



The method of False Position (also called Regula Falsi)

generates approximations in the same manner as the Secant method, but it includes a test to ensure that the root is always bracketed between successive iterations.

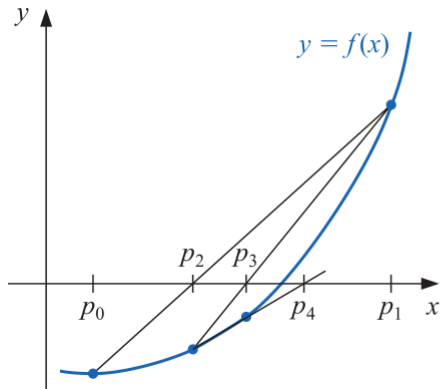
First choose initial approximations p_0 and p_1 with

$$f(p_0)f(p_1) < 0.$$

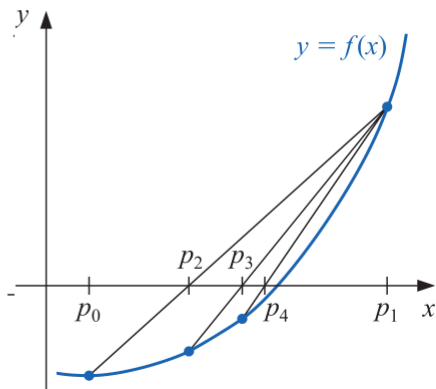
The approximation p_2 is chosen in the same manner as in the Secant method. To decide which secant line to use to compute p_3 ,

- (a) If $f(p_2)f(p_1) < 0$, then p_1 and p_2 bracket a root. Choose p_3 as the x -intercept of the line joining $(p_1, f(p_1))$ and $(p_2, f(p_2))$.
- (b) If not, choose p_3 as the x -intercept of the line joining $(p_0, f(p_0))$ and $(p_2, f(p_2))$, and then interchange the indices on p_0 and p_1 .

Secant Method



Method of False Position



Find a root for $\cos x - x + 2 = 0$ when $x_0 = 1$ and $x_1 = 2$

$$x_k = \frac{x_{k-1}f(x_{k-2}) - x_{k-2}f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})}$$

$$f(x_0) = 1.5403, f(x_1) = -0.41615$$

Secant Method:

$$x_2 = \frac{x_1f(x_0) - x_0f(x_1)}{f(x_0) - f(x_1)}$$

$$x_2 = 1.78729$$

$$x_3 = \frac{x_2f(x_1) - x_1f(x_2)}{f(x_1) - f(x_2)}$$

$$x_3 = 1.78621$$

$$x_4 = \frac{x_3f(x_2) - x_2f(x_3)}{f(x_2) - f(x_3)}$$

$$x_4 = 1.78623$$

$$x_2 = \frac{x_1f(x_0) - x_0f(x_1)}{f(x_0) - f(x_1)}$$

$$x_2 = 1.78729, f(x_2) \cdot f(x_0) < 0$$

$$x_3 = \frac{x_2f(x_0) - x_0f(x_2)}{f(x_0) - f(x_2)}$$

$$x_3 = 1.78622, f(x_3) \cdot f(x_2) < 0$$

$$x_4 = \frac{x_3f(x_2) - x_2f(x_3)}{f(x_2) - f(x_3)}$$

$$x_4 = 1.78623$$

Find a root of the equation $\cos x - x + 2 = 0$ for $x_0 = 1$ and $x_1 = 2$

- ① $4x = \sin x - e^x$
- ② $\cos x - xe^x$
- ③ $x = \cos x, [0.5, \frac{\pi}{4}]$

	False Position	Secant	Newton
n	p_n	p_n	p_n
0	0.5	0.5	0.7853981635
1	0.7853981635	0.7853981635	0.7395361337
2	0.7363841388	0.7363841388	0.7390851781
3	0.7390581392	0.7390581392	0.7390851332
4	0.7390848638	0.7390851493	0.7390851332
5	0.7390851305	0.7390851332	
6	0.7390851332		

A Case Where Bisection Is Preferable to False Position

Use bisection and false position to locate the root of

$$f(x) = x^{10} - 1 \quad \text{between } x = 0 \text{ and } 1.3.$$

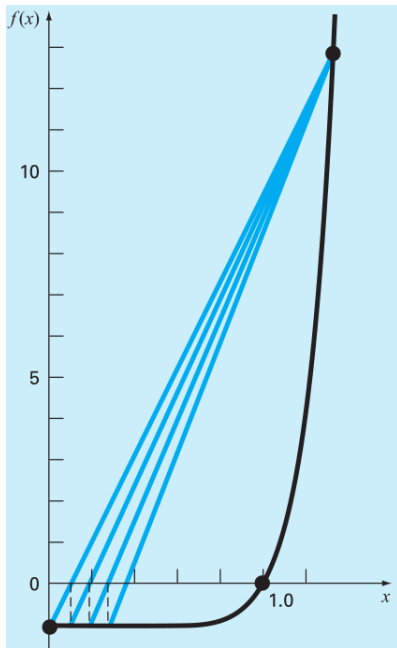
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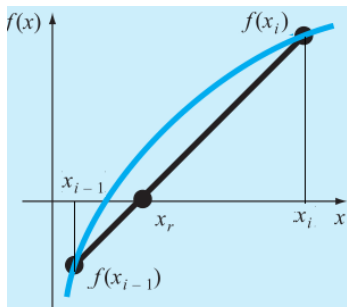
$$f(x) = x^{10} - 1 \quad \text{between } x = 0 \text{ and } 1.3.$$

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.65	100.0	35
2	0.65	1.3	0.975	33.3	2.5
3	0.975	1.3	1.1375	14.3	13.8
4	0.975	1.1375	1.05625	7.7	5.6
5	0.975	1.05625	1.015625	4.0	1.6

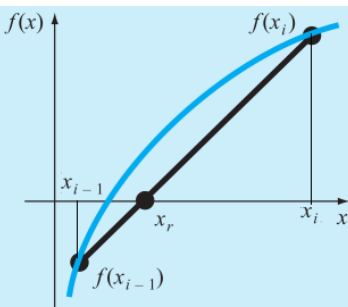
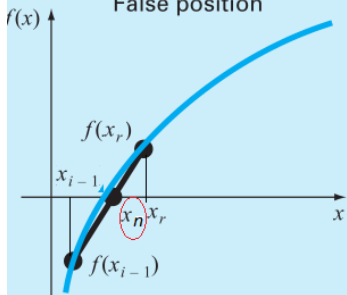
Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.09430		90.6
2	0.09430	1.3	0.18176	48.1	81.8
3	0.18176	1.3	0.26287	30.9	73.7
4	0.26287	1.3	0.33811	22.3	66.2
5	0.33811	1.3	0.40788	17.1	59.2



Use the false-position and secant methods to estimate the root of $f(x) = \ln x$. Start the computation with values of $x_{i-1} = 0.5$ and $x_i = 5.0$.



False position



Secant

